Solution

This equation is Type I even degree reciprocal equation. Hence it can be rewritten as $x^2\left[\left(x^2+\frac{1}{x^2}\right)-10\left(x+\frac{1}{x}\right)+26\right]=0$ Since $x\neq 0$, we get $\left(x^2+\frac{1}{x^2}\right)-10\left(x+\frac{1}{x}\right)+26=0$

Let $y=x+\frac{1}{x}$. Then, we get

$$\left(y^2-2\right)-10y+26=0 \Rightarrow y^2-10y+24=0 \Rightarrow (y-6)(y-4)=0 \Rightarrow y=6 \text{ or } y=4$$

Case (i)

$$y=6 \Rightarrow x+\frac{1}{x}=6 \Rightarrow x=3+2\sqrt{2}, x=3-2\sqrt{2}.$$

Case (ii)

$$y=4 \Rightarrow x+\frac{1}{x}=4 \Rightarrow x=2+\sqrt{3}, x=2-\sqrt{3}$$

Hence, the roots are $3\pm 2\sqrt{2}$, $2\pm\sqrt{3}$

3.8.3 Non-polynomial Equations

Some non-polynomial equations can be solved using polynomial equations. As an example let us consider the equation $\sqrt{15-2x}=x$. First we note that this is not a polynomial equation. Squaring both sides, we get $x^2+2x-15=0$. We know how to solve this polynomial equation. From the solutions of the polynomial equation, we can analyse the given equation. Clearly 3 and -5 are solutions of $x^2+2x-15=0$. If we adopt the notion of assigning only nonnegative values for $\sqrt{\bullet}$ then $x=3$ is the only solution; if we do not adopt the notion, then we get $x=-5$ is also a solution.

Example 3.29: Find solution, if any, of the equation

$$2\cos^2 x-9\cos x+4=0. \quad \ldots (1)$$

Solution

The left hand side of this equation is not a polynomial in x. But it looks like a polynomial. In fact, we can say that this is a polynomial in $\cos x$. However, we can solve equation (1) by using our knowledge on polynomial equations. If we replace $\cos x$ by y, then we get the polynomial equation $2y^2-9y+4=0$ for which 4 and $\frac{1}{2}$ are solutions.

From this we conclude that x must satisfy $\cos x=4$ or $\cos x=\frac{1}{2}$. But $\cos x=4$ is never possible, if we take $\cos x=\frac{1}{2}$, then we get infinitely many real numbers x satisfying $\cos x=\frac{1}{2}$; in fact, for all $n\in\mathbb{Z}$, $x=2n\pi\pm\frac{\pi}{3}$ are solutions for the given equation (1).

If we repeat the steps by taking the equation $\cos^2 x-9\cos x+20=0$, we observe that this equation has no solution.

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